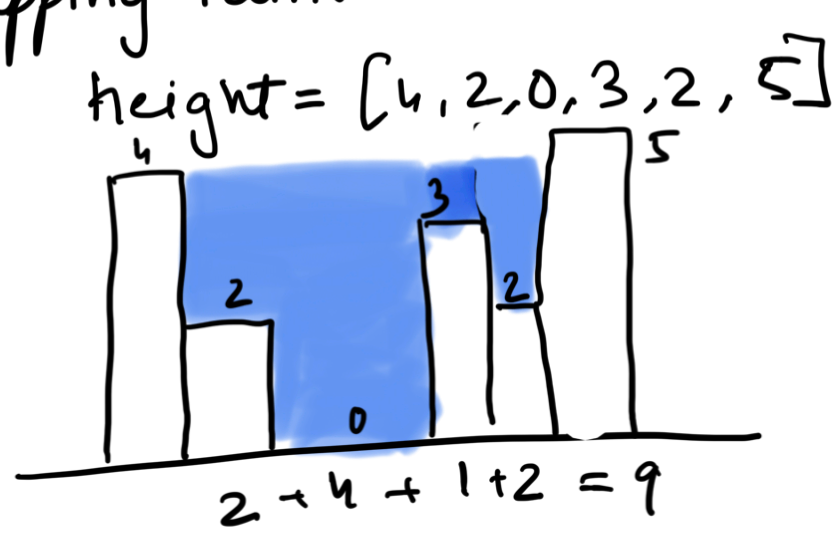


Trapping rainwater



Approach 1 - Prefix array approach

left max [4, 4, 4, 4, 4, 5] TC: O(n)
SC: O(n)

right max [5, 5, 5, 5, 5, 5]

then do in loop and add
 $\min(\text{lmax}[i], \text{rmax}[i]) - \text{ht}[i]$

vector<int> lmax(n, 0)

vector<int> rmax(n, 0)

lmax[0] = height[0]

rmax[n-1] = height[n-1]

for (i = 1; i < n; i++)

lmax[i] = max(lmax[i-1], ht[i])

for (i = n-2; i >= 0; i--)

rmax[i] = max(rmax[i+1], ht[i])

int ans = 0

for (i = 0; i < n; i++)

ans += min(lmax[i], rmax[i]) - ht[i]

return ans

Approach 2 - 2 pointer approach

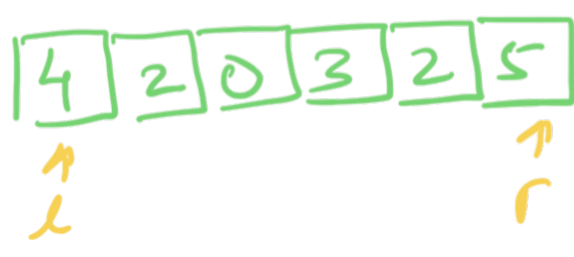
l = 0, r = n-1

TC = O(n)

SC = O(1)

lmax = max(lmax, ht[l]) = 0

rmax = max(rmax, ht[r]) = 0



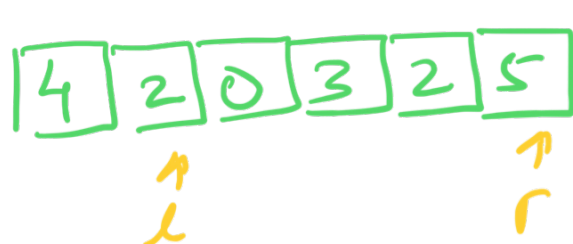
lmax = 4

rmax = 5

lmax < rmax

ans = lmax - ht[l] = 4 - 4 = 0

l++

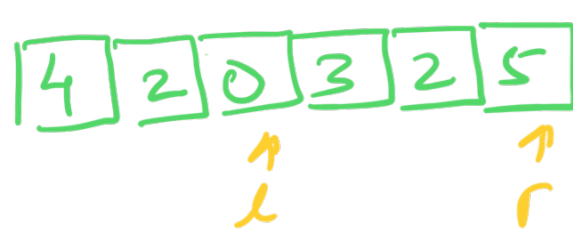


lmax = 4

rmax = 5

lmax < rmax

ans = 0 + 4 - 2 = 2 l++

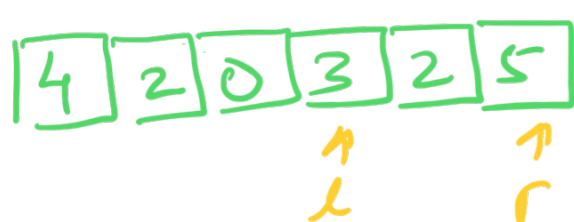


lmax = 4

rmax = 5

lmax < rmax

ans = 2 + 4 - 0 = 6 l++

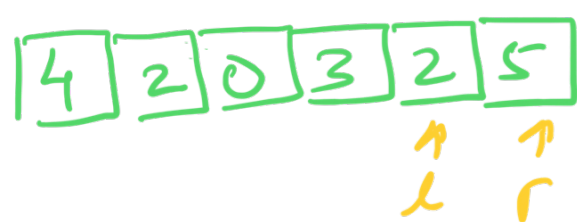


lmax = 4

rmax = 5

lmax < rmax

ans = 6 + 4 - 3 = 7 l++



lmax = 4

rmax = 5

lmax < rmax

ans = 7 + 4 - 2 = 9 l++

l = 0, r = n-1, ans = 0

while (l < r)

lmax = max(lmax, ht[l])

rmax = max(rmax, ht[r])

if (lmax < rmax)

ans += lmax - ht[l]

l++

else

ans += rmax - ht[r]

r--

return ans